THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two lenses of power +3 D and −1 D are in contact. The net power is: 1

(a) +3 D **(b)** +2 D **(c)** +4 D **(d)** −2 D

2. A student states: “A convex lens is called a converging lens.” The correct response is: 1

(a) Correct **(b)** Incorrect — it diverges light
(c) Only for mirrors **(d)** Cannot be determined

3. The refractive index of medium 2 with respect to medium 1 is: 1

(a) $v_1 v_2$ **(b)** $c\, v_1$ **(c)** v_1 / v_2 **(d)** v_2 / v_1

4. An object at 2F of a convex lens gives an image: 1

(a) between F and O **(b)** at F **(c)** at 2F, same size **(d)** at infinity

5. For a magnifying glass, Student A says the object is placed within the focal length; Student B says beyond 2F. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The eye lens is: 1

(a) convex **(b)** plane **(c)** cylindrical **(d)** concave

7. A student states: “The sun appears white at noon because light travels a shorter distance through the atmosphere.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only in winter **(d)** Cannot be determined

8. Which part of the eye has cells sensitive to colour? 1

(a) iris **(b)** lens
(c) rods in the cornea **(d)** cones in the retina

9. The phenomenon of light involved in the formation of a rainbow includes: 1

(a) only scattering
(b) diffraction only
(c) refraction, dispersion and internal reflection
(d) only reflection

10. For a myopic eye, Student A says the image of a distant object forms in front of the retina; Student B says behind. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Define the centre of curvature of a spherical mirror. 2

12. Why is a convex mirror preferred as a rear-view mirror? 2

13. Why do we have two eyes rather than one? 2

14. What happens to the image distance in the eye when we increase the distance of the object? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A concave lens of focal length 10 cm forms an image 6 cm from it. Find the object distance. 3

16. What is the function of the optic nerve and retina? What are rods and cones? 3

17. Explain with a diagram (describe) the refraction of light through a rectangular glass slab. 3

OR

What is the nature of the image formed by a concave mirror if the magnification is 3
(i) +2
(ii) −1
(iii) −0.5?

18. State the causes and correction of presbyopia. 3

OR

Why does a star appear to twinkle? Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A car's side mirror reads “Objects in mirror are closer than they appear”. 5
(i) Name the mirror.
(ii) Why is this warning written?
(iii) If $R = 2$ m, find f.
(iv) For a car 10 m behind, find the image distance.
(v) Is the image real?

OR

(i) What is meant by power of a lens? 5
(ii) Find the power of a combination of a convex lens of focal length 25 cm and a concave lens of focal length 10 cm.
(iii) Is the combination converging?

20. **Case study:** A student held a glass prism in sunlight and saw a band of colours on a wall. 5
(i) Name the phenomenon.
(ii) Which colour is at the top and which at the bottom?
(iii) Why does violet bend most?
(iv) Name the natural phenomenon based on it.
(v) What did Newton prove using a second prism?

OR

(i) What is accommodation? 5
(ii) How do ciliary muscles help in it?
(iii) A myopic person has a far point of 80 cm. Find the power of the corrective lens.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND